

Biology Practice Test 05 (FINAL BOARD SIMULATOR)

Format: Official ISC Board Exam Simulator

Portion: Set 1 Midterm (Comprehensive Syllabus)

Time Allowed: 3 Hours

Maximum Marks: 70

General Instructions:

- This paper simulates the exact format, length, and rigor of the official ISC Board Examination.
 - Answer ALL parts of a question. Read prompts carefully.
 - **NO BLANKS ALLOWED:** A -1 mock penalty applies for leaving any question blank.
 - **Formatting Rule:** You MUST underline genus and species separately when writing scientific names.
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SECTION A (20 Marks)

Answer all questions. Each question carries 1 mark.

Question 1: Name the specific step in the HIV replication cycle where the newly formed virus particles exit the host macrophage, turning it into a viral factory. **Question 2:** Which statistical principle states that allele frequencies in a population remain constant from generation to generation in the absence of evolutionary influences? **Question 3:** Name the type of pollination where pollen grains from one flower are transferred to the stigma of another flower on the *same* plant. **Question 4:** Name the hormone responsible for maintaining the endometrium during pregnancy, primarily secreted by the corpus luteum. **Question 5:** What is the full form of ICSI in the context of Assisted Reproductive Technologies (ART)? **Question 6:** Name the specific type of immunity provided by the transfer of colostrum (containing IgA) from mother to infant. **Question 7:** Name the methanogenic bacterium commonly found in the anaerobic sludge during sewage treatment and in the rumen of cattle. **Question 8:** The flippers of Penguins and Dolphins perform the same function but have different structural origins. What kind of evolution does this represent? **Question 9:** Which genetic disorder is characterized by the presence of a single X chromosome (44A + XO) in females? **Question 10:** Name the physical connection or association of two or more genes located on the same chromosome. **Question 11:** What is the term for the amount of oxygen required by bacteria to oxidize all organic matter in one liter of water? **Question 12:** Name the specific region of the angiosperm ovule where the funicle is attached to the body of the ovule. **Question 13:** Give one example of a genetic trait in humans that exhibits Polygenic Inheritance. **Question 14:** Name the temporary endocrine gland formed in the ovary after ovulation. **Question 15:** Which specific property of normal cells prevents them from growing uncontrollably when they come into contact with other cells? **Question 16:** Name the ecological interaction where one species is harmed and the other is unaffected. **Question 17:** What is the specific role of the enzyme *Integrase* during an HIV infection? **Question 18:** Which phase of population growth is represented by the asymptote in a logistic growth curve? **Question 19:** Name the non-cellular layer surrounding the primary oocyte inside the Graafian follicle. **Question 20:** Give one example of an autoimmune disease in humans.

SECTION B (14 Marks)

Answer all questions. Each question carries 2 marks.

Question 21: Differentiate between ZIFT and GIFT. **Question 22:** Explain the concept of Pleiotropy with one suitable example. **Question 23:** Why are the testes in human males located outside the abdominal cavity in the scrotum? What is the function of the Sertoli cells? **Question 24:** Differentiate between Primary and Secondary sewage treatment based on the principles involved. **Question 25:** Explain how certain animals avoid stressful environmental conditions by "escaping in time" rather than migrating. Give two distinct examples (e.g., bears, snails). **Question 26:** Briefly describe the biological principle behind the use of *Bacillus thuringiensis* (Bt) in agriculture. **Question 27:** Differentiate between Primary and Secondary lymphoid organs with one example of each.

SECTION C (21 Marks)

Answer all questions. Each question carries 3 marks.

Question 28: Describe the structure of a mature pollen grain (microspore). Mention the specific roles of the exine, intine, vegetative cell, and generative cell. **Question 29:** Explain the mechanism of sex determination in honey bees. (Mention ploidy levels). **Question 30:** A couple has a son who is colorblind. The mother is phenotypically normal, and the father has normal vision. Using a Punnett square, explain how this is genetically possible. What is the probability of them having a colorblind daughter? **Question 31:** Describe the three main types of endosperm development in angiosperms (Nuclear, Cellular, Helobial). **Question 32:** What is Artificial Insemination (AI)? Differentiate between AI and Intra Uterine Insemination (IUI). Under what conditions are these techniques recommended? **Question 33:** Draw a well-labeled diagram of a typical Biogas plant. **Question 34:** Discuss the three main types of population age pyramids with respect to expanding, stable, and declining populations.

SECTION D (15 Marks)

Answer any ONE question from each of the three pairs below. Each question carries 5 marks.

Question 35: (Option A) Trace the complete life cycle of the Human Immunodeficiency Virus (HIV) within a human macrophage. Detail the roles of Reverse Transcriptase, Integrase, and the final exit mechanism of the virus. **OR (Option B)** Describe the various innate immunity barriers (physical, physiological, cellular, and cytokine) present in the human body. How does innate immunity fundamentally differ from acquired immunity?

Question 36: (Option A) What is Adaptive Radiation? Explain this evolutionary process taking the classic examples of Darwin's finches in the Galapagos Islands and the Australian marsupials. **OR (Option B)** Explain Morgan's dihybrid cross experiment on *Drosophila melanogaster* regarding eye color and body color. How did his results deviate from Mendel's Law of Independent Assortment, and how did he explain this deviation using the concepts of Linkage and Recombination?

Question 37: (Option A) Trace the development of a typical dicot embryo from the zygote stage to the mature embryo stage. Support your answer with diagrams showing the globular, heart-shaped, and mature stages. **OR (Option B)** Explain the various interactions between species in an ecosystem. Provide a detailed definition and one example for: (i) Mutualism, (ii) Predation, (iii) Parasitism, (iv) Commensalism, and (v) Competition.